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Immigration, Innovation, and Growth

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1 Big picture

- **Question.** Does immigration raise or lower local innovation and wages? The paper asks this in the context of international migration to U.S. counties from 1975 to 2010.
- **Core difficulty.** Migrants do not settle randomly. More innovative or more productive counties may attract more immigrants, so raw correlations between immigration, patenting, and wages are not causal.
- **Main empirical contribution.** The paper builds a new *predicted-ancestry* shift-share instrument. Instead of using realized lagged immigrant shares or realized ancestry, it first instruments the preexisting ancestry distribution itself with historical push-pull migration forces from 1880 onward, then interacts predicted ancestry with contemporaneous national inflows by origin.
- **Main reduced-form result.** Immigration has a positive causal effect on innovation and wages at the five-year horizon. A shock of 10,000 additional immigrants raises the growth of patent flows by about 1.22 patents per 100,000 residents, roughly a 25% increase relative to the mean change in patenting.
- **Wage result.** The same order of magnitude increase in adult immigration raises average annual wages per capita by roughly \$150 over five years, and the gains are larger for more educated natives.
- **Heterogeneity.** More educated immigrants produce much larger gains in both patenting and wages. The wage gains for natives are monotone in native education: almost zero for the least educated and much larger for college and graduate-educated workers.
- **Mechanism.** Immigration raises labor supply, which pushes wages down on impact, but it also expands research labor and stimulates local idea production. The innovation channel dominates at the five-year horizon in the U.S. data.
- **Structural contribution.** A quantitative regional model of endogenous growth and migration is estimated to match the reduced-form elasticity of innovation to immigration. The key estimated elasticity of innovation with respect to research labor is about 0.78.
- **Quantitative implication.** In a counterfactual without the post-1965 increase in U.S. immigration, long-run patenting and output per capita would be about 5% lower.
- **Why the paper matters.** It links immigration to innovation-driven growth, shows why standard shift-share immigration designs can fail under persistent productivity shocks, and provides a concrete way to recover quasi-random local immigration shocks.

2 Data and measurement (key objects)

2.1 Immigration and ancestry

- **Source.** Individual-level IPUMS Census and ACS files: 1880, 1900, 1910, 1920, 1930, 1970, 1980, 1990, 2000, and the 2006–2010 ACS.
- **Coverage.** 3,141 U.S. counties, 195 foreign origin countries, and 10 census waves.

- **Key variables.** $I_{o,d,t}$ is the number of respondents who immigrated from origin country o to county d between $t-1$ and t . $A_{o,d,t}$ is the number of respondents in county d who claim ancestry from origin o at time t .
- **Unit of analysis.** The main reduced-form panel is at the county–five-year level from 1980 to 2010. Table 1 shows roughly 21,987 county-period observations.
- **Why ancestry matters.** Newly arriving migrants tend to settle where same-origin communities already exist. The paper exploits this “social pull” force, but only after purging ancestry of endogenous components.

2.2 Innovation outcomes

- **Source.** USPTO utility-patent microdata, 1975–2010.
- **Assignment.** The baseline assigns patents to counties using assignee locations rather than inventor locations, because most modern patents are owned by corporations.
- **Main innovation measure.** Patent flow in county d at time t is the total number of patents filed in the five-year period ending at t , normalized by 1970 county population.
- **Main dependent variable.** The paper usually studies Δ patent flows per capita between adjacent five-year windows.
- **Important robustness.** Results are similar when patents are assigned by inventor location and when patent quality is weighted by citations.

2.3 Wages and distributional outcomes

- **Source.** County average annual wages come from QCEW, deflated by the PCE price index.
- **Main wage outcome.** The reduced-form wage specification uses the five-year change in average annual real wages.
- **Native nonmovers.** The paper also constructs ten-year wage changes for U.S.-born workers who have not moved counties in the past five years, using IPUMS USA.
- **Education splits.** Native nonmovers are divided into less than high school, high school, some college, college, and graduate education groups.

2.4 Education and spillover measures

- **Adult immigration.** For the education heterogeneity exercises, the endogenous immigration measure is restricted to immigrants age 25 or older.
- **Immigrant education.** The paper computes average years of schooling and average years of college among arriving adult immigrants in each county-period.
- **Spatial spillovers.** Additional measures aggregate immigration shocks from neighboring counties, within-state locations, and concentric distance bands up to 500 km.

2.5 What Table 1 tells you

- **Immigration variation is large.** Non-European immigration averages 1.42 thousand people per county-period, with a standard deviation above 12 thousand.
- **Innovation variation is also large.** The five-year change in patent flows per 100,000 people averages 4.61 with a standard deviation of 37.77.
- **Wage heterogeneity is economically important.** Ten-year wage changes for native non-movers rise sharply with education, making it natural to look at heterogeneous wage effects.
- **Interpretation.** The county-level panel has enough dispersion in immigration, innovation, and wages for a shift-share design to be informative.

3 Models and empirical specifications (core equations + interpretation)

3.1 Baseline reduced-form estimating equation

$$\Delta Y_{d,t} = \delta_t + \delta_{s(d)} + \beta \cdot \text{Immigration}_{d,t} + \epsilon_{d,t}. \quad (1)$$

- **Left-hand side.** $\Delta Y_{d,t}$ is usually the five-year change in patent flows per capita or in real wages.
- **Coefficient of interest.** β is the causal effect of an immigration shock on local innovation or wage growth.
- **Why in differences?** Estimating in changes removes time-invariant county differences in levels of innovation and wages.
- **Additional controls.** The paper often adds county fixed effects, so the identifying variation becomes within-county deviations from trend.

3.2 Step 1: predicting ancestry

$$A_{o,d,t} = \delta_{o,r(d)} + \delta_{c(o),d} + X'_{o,d}\zeta + \sum_{\tau=1880}^t a_{r(d),\tau} I_{o,-r(d),\tau} \frac{I_{\text{Europe},d,\tau}}{I_{\text{Europe},\cdot,\tau}} + v_{o,d,t}. \quad (2)$$

$$\hat{A}_{o,d,t} = \sum_{\tau=1880}^t \hat{a}_{r(d),\tau} \left(I_{o,-r(d),\tau} \frac{I_{\text{Europe},d,\tau}}{I_{\text{Europe},\cdot,\tau}} \right)^{\perp}. \quad (3)$$

- **Economic idea.** Pre-1975 ancestry is predicted using historical *push* factors from each origin country and *pull* factors to each county.
- **Push term.** $I_{o,-r(d),\tau}$ is the number of migrants from origin o settling outside county d 's region at date τ .
- **Pull term.** $I_{\text{Europe},d,\tau}/I_{\text{Europe},\cdot,\tau}$ is county d 's share of European migrants at τ .

- **Why Europe?** The pull factor uses migrants from other origins to proxy county attractiveness while avoiding direct contamination from the origin under study.
- **Why leave-out?** Excluding migrants from the same origin and the same region guards against local and bilateral endogeneity.

3.3 Step 2: predicting immigration from predicted ancestry

$$I_{o,d,t} = \delta_{o,r(d)} + \delta_{c(o),d} + \delta_t + X'_{o,d}\theta + b_t \cdot [\hat{A}_{o,d,t-1} \times \tilde{I}_{o,-r(d),t}] + u_{o,d,t}, \quad (4)$$

where

$$\tilde{I}_{o,-r(d),t} = I_{o,-r(d),t} \left(\frac{I_{\text{Europe},r(d),t}}{I_{\text{Europe},-r(d),t}} \right).$$

Aggregating across origins gives the main county-level immigration instrument:

$$\hat{I}_{.,d,t} = \sum_o \hat{b}_t \cdot [\hat{A}_{o,d,t-1} \times \tilde{I}_{o,-r(d),t}]. \quad (5)$$

- **Interpretation.** This is a shift-share design, but the “share” is predicted ancestry rather than realized immigrant share or realized ancestry.
- **Key advantage.** The instrument only uses variation in ancestry generated by historical push-pull interactions that are plausibly exogenous to current local shocks.
- **Implementation note.** The first-stage coefficients are extremely stable across alternative fixed effects and controls, which is evidence that the instrument is not just picking up obvious confounders.

3.4 Identifying orthogonality condition

$$I_{o,-r(d),\tau} \frac{I_{\text{Europe},d,\tau}}{I_{\text{Europe},\cdot,\tau}} \perp \epsilon_{d,t} \quad \forall o, \tau \leq t. \quad (6)$$

- **Meaning.** Temporary county shocks to innovation and wages after 1975 should not be systematically correlated with the historical push-pull interactions used to predict ancestry.
- **What this blocks.** Both simple reverse causality and more subtle county-origin omitted-factor stories.

3.5 Elasticity specification

$$\text{IHS}(\text{Patents}_{d,t}) = \delta_t + \delta_{s(d)} + \beta_{\text{IHS}} \text{IHS}(\text{Immigration}_{d,t}) + \epsilon_{d,t}. \quad (7)$$

- **Purpose.** The inverse hyperbolic sine specification approximates a log-log regression while allowing zero values.
- **Key estimate.** $\beta_{\text{IHS}} \approx 1.65$, so a 1% immigration increase raises patenting by about 1.6%.
- **Why this matters.** The structural estimation later targets this elasticity to discipline the local idea-production elasticity.

3.6 Structural model: goods production

$$Y_{d,t} = Z_{d,t} Q_{d,t} L_{Y,d,t}^\alpha. \quad (8)$$

- **Interpretation.** Output in county d uses local ideas $Q_{d,t}$ and production labor $L_{Y,d,t}$.
- **Role of $Z_{d,t}$.** County productivity shocks are persistent and are exactly the type of shocks that can invalidate simpler shift-share designs.

3.7 Structural model: idea production

$$N_{d,t} = L_{N,d,t}^\gamma Q_{d,t-1}^{1-\gamma}. \quad (9)$$

- **Interpretation.** New patents are produced by research labor and the existing stock of ideas.
- **Key parameter.** γ is the elasticity of innovation to research labor.
- **Economic meaning.** Immigration can raise innovation by expanding the local supply of researchers.

3.8 Firm problems

$$\max_{N, L_Y} Z_{d,t}(N + Q_{d,t-1})L_Y^\alpha - W_{d,t}L_Y - p_{d,t}N, \quad (10)$$

$$\max_{L_N} p_{d,t}L_N^\gamma Q_{d,t-1}^{1-\gamma} - W_{d,t}L_N. \quad (11)$$

- **Interpretation.** One representative firm produces goods and another produces ideas; both take wages and the local patent price as given.
- **Why useful.** These static firm problems keep the dynamic model tractable while preserving the growth mechanism.

3.9 Migration decisions

$$d = \arg \max_k \mathbb{E}_t \left[W_{k,t+1}^\lambda \left(\frac{A_{o,k,t+1}}{A_{o,\cdot,t+1}} \right)^{1-\lambda} \right] \exp(-\tau_{o,k,t}) \eta_{k,t}(i), \quad (12)$$

$$I_{o,d,t} = I_{o,\cdot,t} \frac{\exp(-\theta \tau_{o,d,t}) \left(\mathbb{E}_t \left[W_{d,t+1}^\lambda \left(\frac{A_{o,d,t+1}}{A_{o,\cdot,t+1}} \right)^{1-\lambda} \right] \right)^\theta}{\sum_{k=1}^D \exp(-\theta \tau_{o,k,t}) \left(\mathbb{E}_t \left[W_{k,t+1}^\lambda \left(\frac{A_{o,k,t+1}}{A_{o,\cdot,t+1}} \right)^{1-\lambda} \right] \right)^\theta}. \quad (13)$$

- **Interpretation.** Migrants choose counties based on expected future wages, the size of same-origin communities, bilateral migration costs, and idiosyncratic tastes.
- **Connection to the empirical design.** The model provides explicit microfoundations for both the “push” and “same-origin community” terms used in the reduced-form instrument.

3.10 Card-style instrument used for comparison

$$\hat{I}_{\cdot,d,t}^{Card} = \sum_{o=1}^O I_{o,\cdot,t} \frac{I_{o,d,t-1}}{I_{o,\cdot,t-1}}. \quad (16)$$

- **Interpretation.** This is the canonical Card (2001) immigration shift-share instrument based on realized lagged immigrant shares.
- **Why the paper criticizes it.** In the model, persistent productivity shocks raise both expected wages and future immigration, so realized lagged shares remain correlated with current productivity and innovation shocks.

4 Identification and instruments (what makes the estimates credible)

4.1 The paper's two main endogeneity stories

- **Reverse causality.** More innovative counties pay higher wages and attract more migrants.
- **County-origin omitted factors.** Some origins may have skills specialized for specific sectors and locations, making ancestry shares correlate with future sectoral innovation shocks.

4.2 Why predicted ancestry is the key innovation

- Standard shift-share designs treat lagged immigrant shares or ancestry as exogenous.
- This paper instead instruments the ancestry distribution itself using historical push and pull factors from 1880 onward.
- The leave-out structure excludes same-origin migrants and local-region migrants in ways designed to purge bilateral and local sorting.

4.3 What Table 2 establishes

- Predicted ancestry times contemporaneous immigration strongly predicts actual immigration.
- The fit is high: R^2 rises from about 0.66 to about 0.71 as controls are added.
- The first-stage coefficients are extremely stable across specifications, which is reassuring because it means the instrument is not fragile to the inclusion of geographic controls, fixed effects, and concurrent European immigration.

4.4 Why the paper claims standard shift-share inference is unreliable

- The paper runs an Adão-Kolesár-Morales randomization test.
- When using predicted ancestry shares, the false rejection rate is about 3.8%, close to the nominal 5%.
- When using realized lagged immigration shares or realized ancestry shares, the false rejection rates jump to about 25–27%.

- Interpretation: conventional shift-share designs are contaminated by persistent local shocks; the predicted-ancestry instrument is not.

4.5 How the structural model strengthens identification

- In the model, persistent productivity shocks raise both wages and immigration, so a Card-style instrument inherits endogeneity.
- Figure 4 shows that realized immigration, the Card IV, and an IV built from realized ancestry are all correlated with productivity shocks or bilateral migration-cost shocks.
- By contrast, the paper’s predicted-ancestry IV is approximately orthogonal to both kinds of shocks.

4.6 Bottom-line identification logic

- **Relevance.** Same-origin community effects make predicted ancestry useful for forecasting where migrants go.
- **Exogeneity.** Historical push-pull interactions and leave-out restrictions isolate quasi-random ancestry variation.
- **Inference.** Randomization tests and the structural model both support the claim that this IV is more credible than conventional shift-share instruments.

5 Tables and figures

5.1 Table 1: Summary statistics

Read:

- Immigration, patent growth, and wage growth all exhibit wide cross-county variation, which is necessary for identification.
- The standard deviation of the immigration shock is sizable relative to the mean, so the instrument creates economically meaningful treatment variation.
- Wage changes among native nonmovers vary strongly by education, foreshadowing heterogeneous wage effects.

5.2 Table 2: Regressions of immigration on push-pull instruments

Read:

- This is the county-origin first stage underlying the immigration shock.
- The interaction of predicted ancestry and national immigration shocks predicts local immigration powerfully in every five-year period from 1980 to 2010.
- Adding distance controls, region-country fixed effects, county-continent fixed effects, and contemporaneous European migration barely changes the coefficients.
- Takeaway: the first stage is both strong and stable.

TABLE 1—SUMMARY STATISTICS BY COUNTY-YEAR

	N	Mean	SD	IQR
<i>Immigration flows and population change</i>				
Immigration _{it}	21,987	1.42	12.21	0.22
Δ Population _{it}	21,986	4.02	19.64	2.54
Immigration Shock ($\hat{I}_{it}$)	21,987	-0.00	4.99	0.24
<i>Patents</i>				
Patent flows per 100,000 people	21,987	31.27	85.21	22.08
5-year difference in patent flows (PF) per 100,000 people (assignee)	18,846	4.61	37.77	6.35
5-year difference in PF per 100,000 people (assignee, citation weighted)	18,846	4.02	50.09	5.71
5-year difference in PF per 100,000 people (inventor)	18,846	8.55	46.93	16.99
5-year difference in PF per 100,000 people (inventor, citation weighted)	18,846	8.02	72.53	16.12
<i>Wages</i>				
5-year difference in average annual wage	21,977	18.93	56.52	25.80
10-year difference in average annual wage of native nonmovers aged 25+ (NNM)	6,274	15.00	30.81	37.84
10-year difference in average annual wage of NNM with less than high school	6,274	-6.36	36.51	41.75
10-year difference in average annual wage of NNM with high school	6,274	-1.90	29.86	42.36
10-year difference in average annual wage of NNM with some college	6,274	7.54	32.60	43.81
10-year difference in average annual wage of NNM with B.A.	6,274	22.46	48.31	58.26
10-year difference in average annual wage of NNM with graduate school	6,274	46.97	74.06	96.64
<i>Immigration and education</i>				
Immigration _{it} (Age 25+)	21,987	0.80	6.91	0.11
Average Years College _{it} (Age 25+)	21,987	1.50	1.41	1.82
Average Years Education _{it} (Age 25+)	21,987	10.88	3.65	4.59
<i>Spillovers</i>				
Immigration _{it,d,t}	21,987	114.21	216.16	84.90
Neighbors' Immigration _{it,d,t} (inverse distance weight)	21,987	1.15	0.78	0.65
Immigration _{100mi,d,t} (other counties within 100 km)	21,987	18.58	64.65	9.21
Immigration _{250mi,d,t} (other counties within 250 km)	21,987	74.96	135.50	67.60
Immigration _{500mi,d,t} (other counties within 500 km)	21,987	123.10	149.52	143.69

Notes: This table displays the number of observations, mean, standard deviation, and interquartile range for all outcome variables considered, as well as the variables for immigration and the immigration instrument. The first section of the table contains summary statistics for immigration (here we focus only on non-European migration) and population growth in 1,000s of people. The second section lists summary statistics for patenting and differences in patenting per 100,000 people. The third section reports summary statistics for wages (\$100). Finally, the fourth and fifth section provide summary statistics on the immigration variables used in the education and spillovers analyses, respectively. Variables for immigration, population growth, and education are all for five-year periods, as are the differenced outcomes except in the case of differences in average annual wage for natives and native nonmovers, which are over 10-year periods.

TABLE 2—REGRESSIONS OF IMMIGRATION ON PUSH-PULL INSTRUMENTS AT THE COUNTRY-COUNTY LEVEL

	Immigration _{it,d}				
	(1)	(2)	(3)	(4)	(5)
$\hat{A}_{it,d,1975} \times \hat{I}_{it,-r(d),1980}$	0.0036 (0.0000)	0.0036 (0.0000)	0.0035 (0.0000)	0.0035 (0.0000)	0.0035 (0.0000)
$\hat{A}_{it,d,1980} \times \hat{I}_{it,-r(d),1985}$	0.0016 (0.0000)	0.0016 (0.0000)	0.0016 (0.0000)	0.0016 (0.0000)	0.0016 (0.0000)
$\hat{A}_{it,d,1985} \times \hat{I}_{it,-r(d),1990}$	0.0018 (0.0000)	0.0018 (0.0000)	0.0018 (0.0000)	0.0018 (0.0000)	0.0018 (0.0000)
$\hat{A}_{it,d,1990} \times \hat{I}_{it,-r(d),1995}$	0.0005 (0.0000)	0.0005 (0.0000)	0.0005 (0.0000)	0.0005 (0.0000)	0.0005 (0.0000)
$\hat{A}_{it,d,1995} \times \hat{I}_{it,-r(d),2000}$	0.0004 (0.0000)	0.0004 (0.0000)	0.0004 (0.0000)	0.0004 (0.0000)	0.0004 (0.0000)
$\hat{A}_{it,d,2000} \times \hat{I}_{it,-r(d),2005}$	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)
$\hat{A}_{it,d,2005} \times \hat{I}_{it,-r(d),2010}$	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)	0.0002 (0.0000)
$I_{Europe,it}$				0.0109 (0.0031)	
$\hat{I}_{Europe,it}^{\text{EU}}$					0.3913 (0.1558)
Observations	3,583,881	3,583,881	3,583,881	3,583,881	3,583,881
R ²	0.656	0.657	0.709	0.709	0.709
Distance	No	Yes	Yes	Yes	Yes
Latitude dis.	No	Yes	Yes	Yes	Yes
Region-country fixed effects	No	No	Yes	Yes	Yes
County-continent fixed effects	No	No	Yes	Yes	Yes
Time fixed effects	No	No	Yes	Yes	Yes
Concurrent European immigration	No	No	No	Yes	No
Contemporaneous push-pull	No	No	No	No	Yes

Notes: This table reports coefficient estimates for step 2 of our instrument construction, shown in equation (4), at the country-county level. Interpretation: in column 1, Immigration_{it,d}^{EU} loads on $\hat{A}_{it,d,1975} \times \hat{I}_{it,-r(d),1980}$ with a coefficient 0.0036 while Immigration_{it,d,1985} loads on $\hat{A}_{it,d,1980} \times \hat{I}_{it,-r(d),1985}$ with a coefficient 0.0016. Moving from column 1 to column 3 we introduce controls for distance and latitude distance and then fixed effects into the regression specification. Column 4 adds contemporaneous European migration as a control while column 5 instead introduces the contemporaneous push-economic pull factor for non-European migration. Standard errors are clustered by country for all specifications.

5.3 Table 3: Immigration and patenting

Read:

- This is the main causal result of the paper.
- The IV coefficient of about 0.122 means 10,000 more immigrants increase patent growth by 1.22 patents per 100,000 people over five years.
- OLS is larger and less stable, which is consistent with positive sorting of migrants toward innovative places.
- The IHS estimate of about 1.652 is the key elasticity targeted in the structural estimation.

5.4 Table 4: Immigration and wages

Read:

- Immigration increases local wages rather than lowering them at the five-year horizon.
- For native nonmovers, the gains are extremely uneven across education groups: close to zero for the least educated, positive and increasing for high school, some college, college, and graduate workers.
- This table is central for the paper's claim that immigration raises local inequality even while lifting average wages.

5.5 Table 5: Education analysis

Read:

- More educated immigrants generate substantially larger gains in both innovation and wages.

TABLE 3—COUNTY-LEVEL PANEL REGRESSIONS OF DIFFERENCE IN PATENTING ON IMMIGRATION

	Δ^{5yr} Patent flows per capita			IHS(Patent flows per capita)
	(1)	(2)	(3)	
<i>Panel A. OLS</i>				
<i>Immigration_{it}</i>	0.200 (0.096)	0.194 (0.096)	0.309 (0.197)	
<i>IHS(Immigration_{it})</i>				1.751 (0.140)
Observations	18,846	18,840	18,846	21,987
R^2	0.030	0.053	0.190	0.577
<i>Panel B. IV</i>				
<i>Immigration_{it}</i>	0.122 (0.045)	0.115 (0.045)	0.181 (0.087)	
<i>IHS(Immigration_{it})</i>				1.652 (0.150)
Observations	18,846	18,840	18,846	21,987
First stage F -stat	911	807	85	94
AR wald F -test p -value	0.014	0.021	0.013	0.000
<i>Panel C. First stage</i>		<i>Immigration_{it}^t</i>		<i>IHS(Immigration_{it}^t)</i>
<i>Immigration shock ($\hat{I}_{it}$)</i>	2.119 (0.070)	2.124 (0.075)	1.610 (0.175)	
<i>IHS($\hat{I}_{it}$)</i>				0.792 (0.081)
Observations	18,846	18,840	18,846	21,987
R^2	0.762	0.766	0.956	0.541
Geography fixed effects	State	State	County	State
Time fixed effects	Yes	Yes	Yes	Yes
State-time fixed effects	No	Yes	No	No

Notes: Panels A and B of this table reports the OLS and IV results, respectively, of the estimation of equation (1) where the dependent variable is the change in patenting per 100,000 people (population is based on baseline 1970 levels) in county d in the five-year period ending in t and the endogenous variable is non-European immigration (1,000s) in d and period t in columns 1 to 3; column 4 reports results for a comparable regression of the inverse-hyperbolic sine (IHS) of patenting per 100,000 people on the IHS of non-European immigration (1,000s). Panel C reports the results for step 3 of instrument construction, or the coefficient estimates for the first-stage specification for non-European immigration (1,000s) for the instrument described in equation (5). The table includes the first-stage F -statistic on the excluded instrument and the p -value for the Anderson-Rubin Wald F -test for each of the IV specifications. Standard errors are clustered by state for all specifications.

TABLE 4—IMMIGRATION AND WAGES

	Δ^{5yr} wages		Δ^{10yr} wages					
	Average annual wage		All native nonmovers	Less than high school	High school	1 to 3 years of college	4 years of college	5+ years of college
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Immigration_{it}</i>	0.149 (0.030)	0.217 (0.098)	0.108 (0.034)	-0.007 (0.007)	0.017 (0.005)	0.029 (0.011)	0.085 (0.025)	0.247 (0.085)
Observations	21,977	21,976	6,274	6,274	6,274	6,274	6,274	6,274
First stage F -stat	903	37	936	936	936	936	936	936
AR wald F -test p -value	0.000	0.039	0.006	0.323	0.001	0.021	0.003	0.010
Geography fixed effects	State	County	State	State	State	State	State	State
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the results of our IV specification, described in equation (1), for each of our dependent variables with non-European immigration for those aged 25+ (1,000s) to d in t as the endogenous variable. Column 1 reports the results of our IV regression where the dependent variable is the change in the average annual real wage (\$100s, at 2010 prices) over the five-year period ending in t . Column 2 repeats the specification of column 1 but adds county fixed effects. Columns 3 through 8 report results where the dependent variable is the change in the average annual real wage (\$100s, at 2010 prices) for native nonmovers (those with birthplace listed as a US state or Washington, DC who have not moved in the last 5 years) over the 10-year period ending in t on instrumented non-European immigration for the 10-year period ending in t . Column 3 relies on wage data for all native nonmovers aged 25 and older while columns 4 through 8 further limit the sample of wage data to native nonmovers aged 25 and older with education levels of less than high school, high school, 1 to 3 years of college, 4 years of college, and at least 5 years of college, respectively. We report the first-stage F -statistic on the excluded instrument and the p -value for the Anderson-Rubin Wald F -test for each specification. Standard errors are clustered by state for all specifications.

- The interaction terms with years of education and years of college are positive and economically large.
- The nonparametric tercile split tells the same story: highly educated immigrant inflows have one-order-of-magnitude larger effects than average inflows.
- This table links the innovation channel to immigrant skill composition.

5.6 Figure 1: The impact of an immigration shock

Read:

- The figure shows the structural model's impulse responses.
- Immigration has an immediate effect on labor supply, a persistent positive effect on patenting, and a wage response that depends on the elasticity of idea production with respect to research labor.
- With the estimated γ , the positive innovation channel dominates the negative labor-supply channel at the five-year horizon.

5.7 Table 6: Parameters and model fit

Read:

- The model matches the targeted reduced-form elasticity of innovation to immigration very closely (data 1.652 versus model 1.641).
- The estimated elasticity of patenting to research labor is $\gamma \approx 0.781$.

TABLE 5—EDUCATION ANALYSIS

	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Δ^{5yr} patent flows per capita</i>					
$Immigration_{d,t}$	0.212 (0.081)	0.254 (0.082)	0.584 (0.356)	0.514 (0.128)	
$Average\ Years\ Education_{d,t} \times Immigration_{d,t}$		0.281 (0.094)	0.280 (0.128)		
$Average\ Years\ College_{d,t} \times Immigration_{d,t}$				1.076 (0.283)	
$1\{Low\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					-1.671 (5.620)
$1\{Medium\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					0.105 (0.062)
$1\{High\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					1.705 (0.830)
Observations	18,846	18,846	18,846	18,846	18,846
Montiel-Pflueger effective F -stat	39	38; 13	18; 15	19; 5	4; 37; 3
AR Wald F -test p -value	0.000	0.000	0.000	0.000	0.000
<i>Panel B. Δ^{5yr} wages</i>					
$Immigration_{d,t}$	0.243 (0.095)	0.298 (0.058)	0.761 (0.385)	0.424 (0.093)	
$Average\ Years\ Education_{d,t} \times Immigration_{d,t}$		0.251 (0.055)	0.238 (0.097)		
$Average\ Years\ College_{d,t} \times Immigration_{d,t}$				0.640 (0.101)	
$1\{Low\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					-0.264 (0.259)
$1\{Medium\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					0.183 (0.064)
$1\{High\ Avg.\ Years\ Education\} \times Immigration_{d,t}$					1.637 (0.360)
Observations	21,977	21,977	21,976	21,977	21,977
Montiel-Pflueger effective F -stat	42	39; 16	15; 18	19; 5	41; 29; 3
AR Wald F -test p -value	0.000	0.000	0.000	0.000	0.000
Geography fixed effects	State	State	County	State	State
Time fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the results of our IV specification (1) for the change in patenting per 100,000 people (population is based on baseline 1970 levels) in panel A and the 5-year difference in county-level average real annual wages (\$100s, at 2010 prices) in panel B. Column 1 repeats our main specification but adjusting the migrant pool to those aged 25+ (1,000s). Columns 2 and 3 then add a second endogenous variable for the interaction of immigration with the (demeaned) average years of education of the migrants arriving in the destination county, whereas column 4 adds (demeaned) average years of college education of those migrants. Repeating the regression in column 2 of the second panel for the 10-year difference in average annual wages (\$100s, at 2010 prices) of native nonmovers (US-born working individuals who have not moved outside of the county within the past 5 years) on 10-year migration and corresponding education results in coefficients of 0.249 (0.054) and 0.138 (0.036) on immigration and average years of education times immigration, respectively. Column 5 uses as endogenous variables adult immigration interacted with indicators for the terciles of average years of education of migrants across counties in period t . In all specifications, for instrumentation, we exploit the fact that in our initial instrument construction we created quasi-exogenous immigration shocks for each origin country- $o \times$ destination county- d pair in each time period t ; each specification utilizes the predicted immigration shocks for each of the top 20 origin nations as a joint set of instruments. For each regression we report the Montiel Olea and Pflueger (2013) effective F -statistic. In regressions with multiple endogenous variables we use the orthogonalization method described in Angrist and Pischke (2009, p. 217–218). See the main text for details. We report the p -value for the Anderson-Rubin Wald F -test for each specification. Standard errors are clustered by state for all specifications.

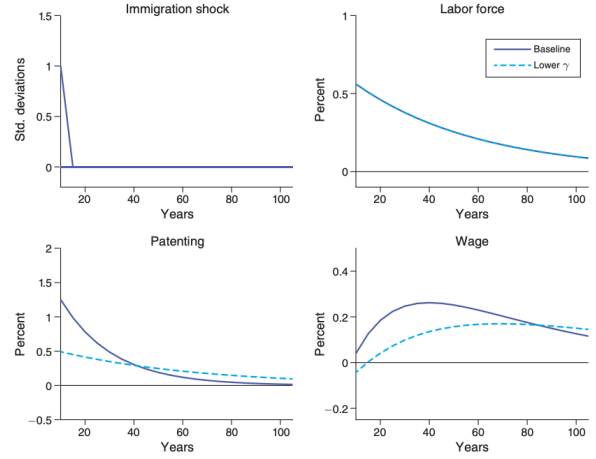


FIGURE 1. THE IMPACT OF AN IMMIGRATION SHOCK

Notes: The figure plots impulse response functions to a one-standard deviation immigration shock in period 1. The top left panel plots the immigration shock $\nu_{o,d,t}$. The top right panel plots the labor force $l_{d,t}$. The bottom left panel plots the response of the wage $w_{d,t}$. The immigration shock is from a single origin o , and the responses of the labor force, patenting, and the wage are local responses for a county d . The baseline impacts are in solid blue, while the lighter dashed line lowers the parameter γ from its baseline of 0.781 to 0.5.

- County productivity shocks are highly persistent, which is why a conventional Card-style IV is dangerous in this setting.

TABLE 6—PARAMETERS AND MODEL FIT		
Panel A. Moments		
	Data	Model
IV coeff., $patenting_{d,t}$ on immigration $I_{d,t}$	1.6519 (0.1500)	1.6410
Standard deviation, o immigration $I_{o,t}$	0.4061 (0.0284)	0.3975
Standard deviation d immigration $I_{d,t}$	0.1794 (0.0110)	0.1655
Standard deviation, o - d immigration $I_{o,d,t}$	0.0716 (0.0117)	0.1138
Autocorrelation, output per capita $Y_{d,t}/L_{d,t}$	0.9611 (0.0057)	0.9646
Autocorrelation, $patenting_{d,t}$	0.9309 (0.0065)	0.8925
Panel B. Estimated parameters		
	Symbol	Value
Elasticity, patenting to labor	γ	0.7807 (0.0857)
Autocorrelation, county TFP	ρ	0.8631 (0.0230)
Standard deviation, county TFP shocks	σ_ϵ	0.0203 (0.0090)
Standard deviation, immigration push shocks	σ_ν	0.5951 (0.0793)
Standard deviation, bilateral immigration shocks	σ_τ	0.5200 (0.0707)

Notes: The top panel A reports targeted data moments versus simulated model moments. The bottom panel B reports the estimated parameters. The standard errors, in parentheses beneath moments and estimates, are clustered by state.

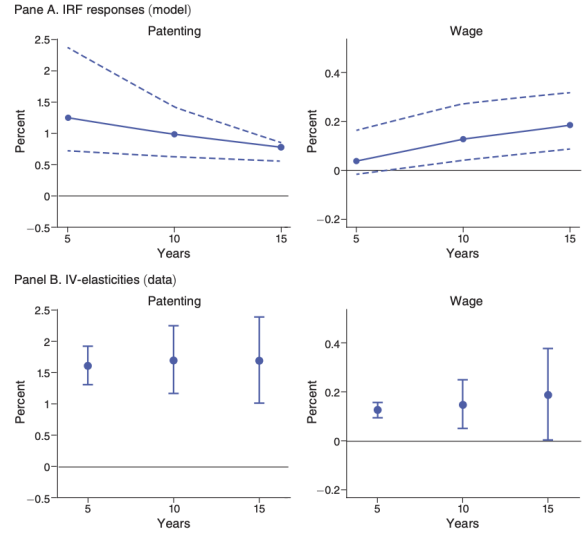


FIGURE 2. IMMIGRATION, INNOVATION, AND WAGES: MODEL VERSUS DATA

Notes: Panel A plots model-estimated impulse response functions for patenting (left figure) and wages (right figure) to a one-standard deviation immigration shock in period 1 (with dashed lines representing standard error boundaries associated with γ). Panel B displays the results of estimating equation (1), where the endogenous variable is the inverse hyperbolic (IHS) of non-European immigration (1,000s) to county d at time t and the dependent variable is the IHS of 5-year patent flows per 100,000 people (left figure) or IHS of wages (right figure). The figures plot the coefficient estimate and 95 percent confidence intervals on the endogenous variable for separate regressions where the outcome is measured in period t , $t+1$, and $t+2$ (for the latter two regressions, we include controls for the immigration shock in $t+1$ and the immigration shocks in $t+1$ and $t+2$, respectively).

5.8 Figure 2: Immigration, innovation, and wages — model versus data

Read:

- Panel A shows the model implies persistent patent and wage responses to an immigration shock.
- Panel B shows the reduced-form data display the same qualitative pattern at 5-, 10-, and 15-year horizons.
- This is important because the model is only directly targeted to the contemporaneous patent response, not to the wage path.

5.9 Figure 3: Removing the post-INA immigration increase

Read:

- The counterfactual reduces long-run labor-force growth by roughly one-sixth relative to realized post-1965 history.
- Patenting per capita falls quickly and remains lower; output per capita declines more gradually as the stock of ideas erodes.
- The long-run output loss is about 5%, which is the paper's headline macro implication.

TABLE 3—COUNTY-LEVEL PANEL REGRESSIONS OF DIFFERENCE IN PATENTING ON IMMIGRATION

	Δ^{5yr} Patent flows per capita			IHS(Patent flows per capita)
	(1)	(2)	(3)	
<i>Panel A. OLS</i>				
<i>Immigration_{it}</i>	0.200 (0.096)	0.194 (0.096)	0.309 (0.197)	
<i>IHS(Immigration_{it})</i>				1.751 (0.140)
Observations	18,846	18,840	18,846	21,987
R^2	0.030	0.053	0.190	0.577
<i>Panel B. IV</i>				
<i>Immigration_{it}</i>	0.122 (0.045)	0.115 (0.045)	0.181 (0.087)	
<i>IHS(Immigration_{it})</i>				1.652 (0.150)
Observations	18,846	18,840	18,846	21,987
First stage F -stat	911	807	85	94
AR wald F -test p -value	0.014	0.021	0.013	0.000
<i>Panel C. First stage</i>		<i>Immigration_{it}^t</i>		<i>IHS(Immigration_{it})^t</i>
<i>Immigration shock ($\hat{I}_{it}$)</i>	2.119 (0.070)	2.124 (0.075)	1.610 (0.175)	
<i>IHS($\hat{I}_{it}$)</i>				0.792 (0.081)
Observations	18,846	18,840	18,846	21,987
R^2	0.762	0.766	0.956	0.541
Geography fixed effects	State	State	County	State
Time fixed effects	Yes	Yes	Yes	Yes
State-time fixed effects	No	Yes	No	No

Notes: Panels A and B of this table reports the OLS and IV results, respectively, of the estimation of equation (1) where the dependent variable is the change in patenting per 100,000 people (population is based on baseline 1970 levels) in county d in the five-year period ending in t and the endogenous variable is non-European immigration (1,000s) in d and period t in columns 1 to 3; column 4 reports results for a comparable regression of the inverse-hyperbolic sine (IHS) of patenting per 100,000 people on the IHS of non-European immigration (1,000s). Panel C reports the results for step 3 of instrument construction, or the coefficient estimates for the first-stage specification for non-European immigration (1,000s) for the instrument described in equation (5). The table includes the first-stage F -statistic on the excluded instrument and the p -value for the Anderson-Rubin Wald F -test for each of the IV specifications. Standard errors are clustered by state for all specifications.

TABLE 4—IMMIGRATION AND WAGES

	Δ^{5yr} wages		Δ^{10yr} wages					
	Average annual wage		All native nonmovers	Less than high school	High school	1 to 3 years of college	4 years of college	5+ years of college
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Immigration_{it}</i>	0.149 (0.030)	0.217 (0.098)	0.108 (0.034)	-0.007 (0.007)	0.017 (0.005)	0.029 (0.011)	0.085 (0.025)	0.247 (0.085)
Observations	21,977	21,976	6,274	6,274	6,274	6,274	6,274	6,274
First stage F -stat	903	37	936	936	936	936	936	936
AR wald F -test p -value	0.000	0.039	0.006	0.323	0.001	0.021	0.003	0.010
Geography fixed effects	State	County	State	State	State	State	State	State
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the results of our IV specification, described in equation (1), for each of our dependent variables with non-European immigration for those aged 25+ (1,000s) to d in t as the endogenous variable. Column 1 reports the results of our IV regression where the dependent variable is the change in the average annual real wage (\$100s, at 2010 prices) over the five-year period ending in t . Column 2 repeats the specification of column 1 but adds county fixed effects. Columns 3 through 8 report results where the dependent variable is the change in the average annual real wage (\$100s, at 2010 prices) for native nonmovers (those with birthplace listed as a US state or Washington, DC who have not moved in the last 5 years) over the 10-year period ending in t on instrumented non-European immigration for the 10-year period ending in t . Column 3 relies on wage data for all native nonmovers aged 25 and older while columns 4 through 8 further limit the sample of wage data to native nonmovers aged 25 and older with education levels of less than high school, high school, 1 to 3 years of college, 4 years of college, and at least 5 years of college, respectively. We report the first-stage F -statistic on the excluded instrument and the p -value for the Anderson-Rubin Wald F -test for each specification. Standard errors are clustered by state for all specifications.

5.10 Figure 4: TFP, bilateral shocks, and instrumental variables

Read:

- The left panel shows the Card IV and the realized-ancestry IV remain correlated with persistent productivity shocks.
- The right panel shows those same IVs also correlate with bilateral migration-cost shocks.
- The predicted-ancestry IV is close to orthogonal to both, which is exactly the paper's identification claim.

6 Short summary

- **Conclusion.** Immigration raises local innovation and wages in U.S. counties once one isolates plausibly exogenous immigration shocks.
- **Intuition.** Immigration adds researchers and expands local idea production; that innovation channel more than offsets the short-run downward pressure of labor supply on wages.
- **Empirical lesson.** Standard shift-share immigration IVs can fail in the presence of persistent productivity shocks; instrumenting ancestry itself is a powerful fix.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Immigration has a positive causal effect on patenting and wage growth at the local level.
- **Magnitude.** 10,000 more immigrants raise five-year patent growth by about 1.22 patents per 100,000 residents and raise annual wages by roughly \$150.
- **Credibility.** The estimates rest on a historical predicted-ancestry shift-share instrument, strong first stages, randomization tests, and a structural model showing why Card-style IVs are contaminated.
- **Mechanism evidence.** Effects are stronger when immigrants are more educated; most of the innovation response comes from domestic inventors; spillovers extend to nearby counties.
- **Quantitative implication.** The estimated local idea-production elasticity is large enough that post-1965 immigration plausibly increased long-run U.S. patenting and output per capita by around 5%.
- **Broader message.** Temporary labor-supply shocks can have persistent growth effects when they change the local production of ideas.

7.2 Economic intuition

- Immigration is not only a labor-supply shock; it is also a shock to the size of the local research sector.
- In places where idea production has sizeable local scale effects, more researchers generate more patents, which lift productivity and wages over time.
- The dynamic innovation channel helps reconcile why some studies find immigration depressing wages in the short run while others find positive longer-run effects.
- Because migrants choose destinations endogenously, credible identification requires purging local immigrant shares of the same persistent productivity shocks that also raise innovation.

7.3 Takeaway

This is a very strong paper on both identification and mechanism. Empirically, it shows that immigration causes more local innovation and, over a five-year horizon, higher wages. Methodologically, its main lesson is that one should not take preexisting immigrant shares as exogenous just because they are lagged; if productivity shocks are persistent, they remain contaminated for a long time. Quantitatively, the paper argues that immigration matters for growth not mainly because it adds workers today, but because it expands the local creation of ideas that raise productivity tomorrow.